Criterion C: System Overview

1. Determine updates for user input
2. Determine which screen to display
3. Update and display proper objects
4. Display UI elements

1. First, the program checks for user input, such as keyboard presses and mouse inputs. For both mouse and keyboard inputs, some inputs cause an immediate change, while other inputs are simply stored for later use. Things that are handled immediately are things such as the dropdown select, object selection, and updating booleans based on keyboard inputs.
2. Based on some booleans, the program determines which screen it is on. The program has a few screens it can swap between. The main menu, which is also the start screen. The simulation, which is where objects are displayed and potentially updated. The paused screen for the simulation, and the settings screen, where some visual settings can be adjusted.
3. On each screen, the stored inputs from step 1 are now resolved. In addition, if the simulation is currently selected, and the simulation is not paused, the physics for the objects will be updated. Regardless of if the simulation is paused, all the objects will also be displayed.
4. After the 3d rendering is complete, all 2d aspects of the simulation are rendered. This includes buttons, dropdowns, textboxes, and some other menu aspects. While not on the simulation, all rendering is 2 dimensionally.